7.

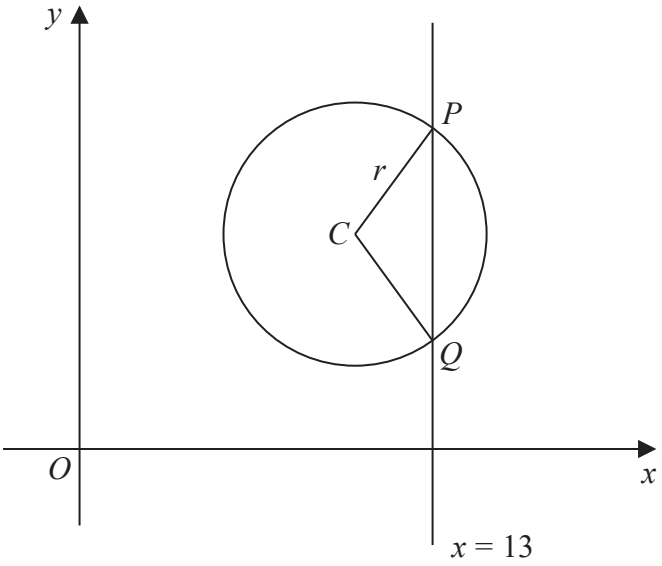



**Figure 1**

The circle with equation

$$x^2 + y^2 - 20x - 16y + 139 = 0$$

had centre $C$ and radius $r$.

(a) Find the coordinates of $C$.

**(2)**

(b) Show that $r = 5$

**(2)**

The line with equation $x = 13$ crosses the circle at the points $P$ and $Q$ as shown in Figure 1.

(c) Find the $y$ coordinate of $P$ and the $y$ coordinate of $Q$.

**(3)**

A tangent to the circle from $O$ touches the circle at point $X$.

(d) Find, in surd form, the length $OX$.

**(3)**